GIT-207

Practical 2

Microbiology & Immunology Department

By

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Objectives:

By the end of this session, the student will be able to:

- List the main viruses causing hepatitis
- Demonstrate the laboratory diagnosis and how to analyze results of HBV

Laboratory diagnosis of viral hepatitis

Viral hepatitis can be caused by:

HAV, HEV, HBV, HCV & HDV

Diagnosis of viral hepatitis is performed by:

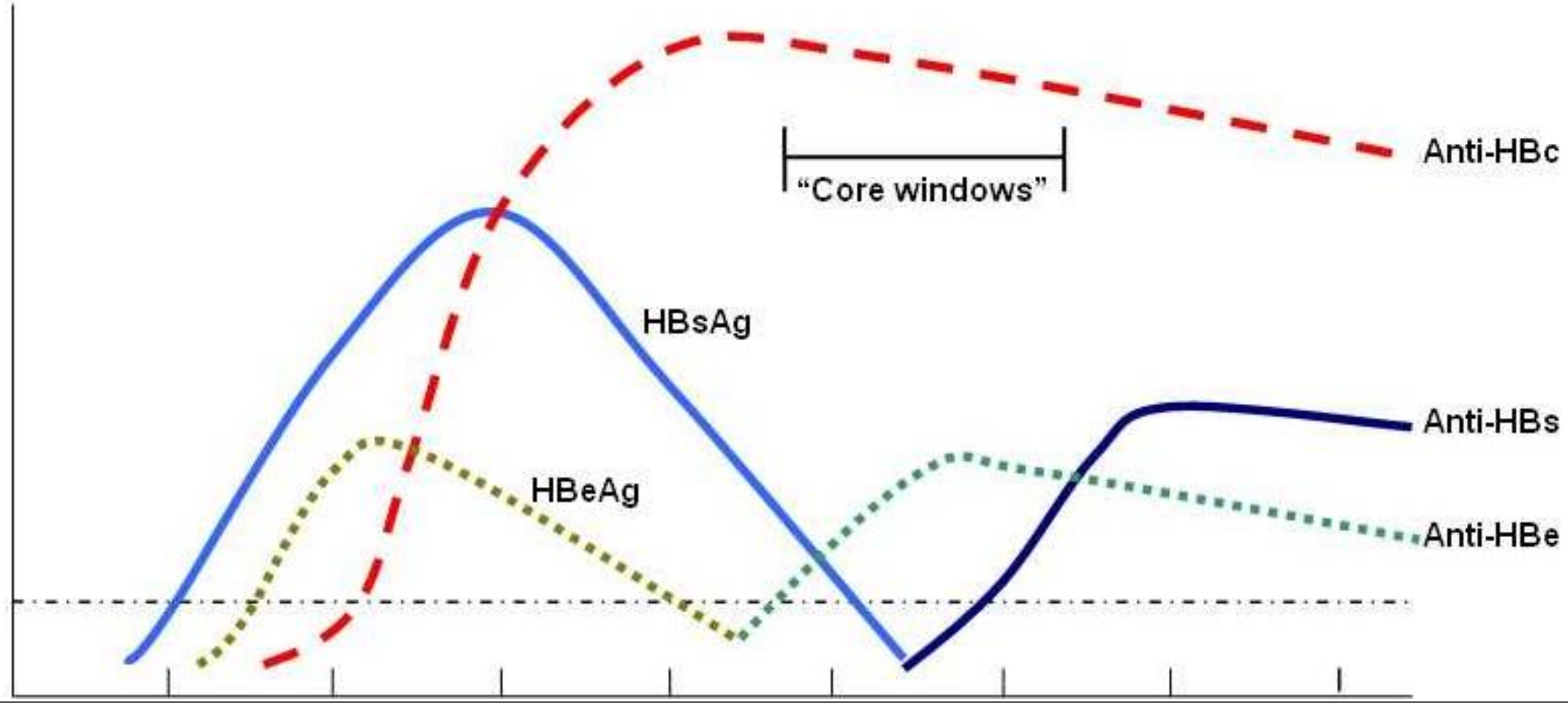
- **Liver function testing:** elevated liver enzymes (AST & ALT), elevated bilirubin…
- **Serology**: to detect antiviral antibodies (IgM & IgG) by ELISA
- **PCR:** to detect viral genome

Hepatitis B markers

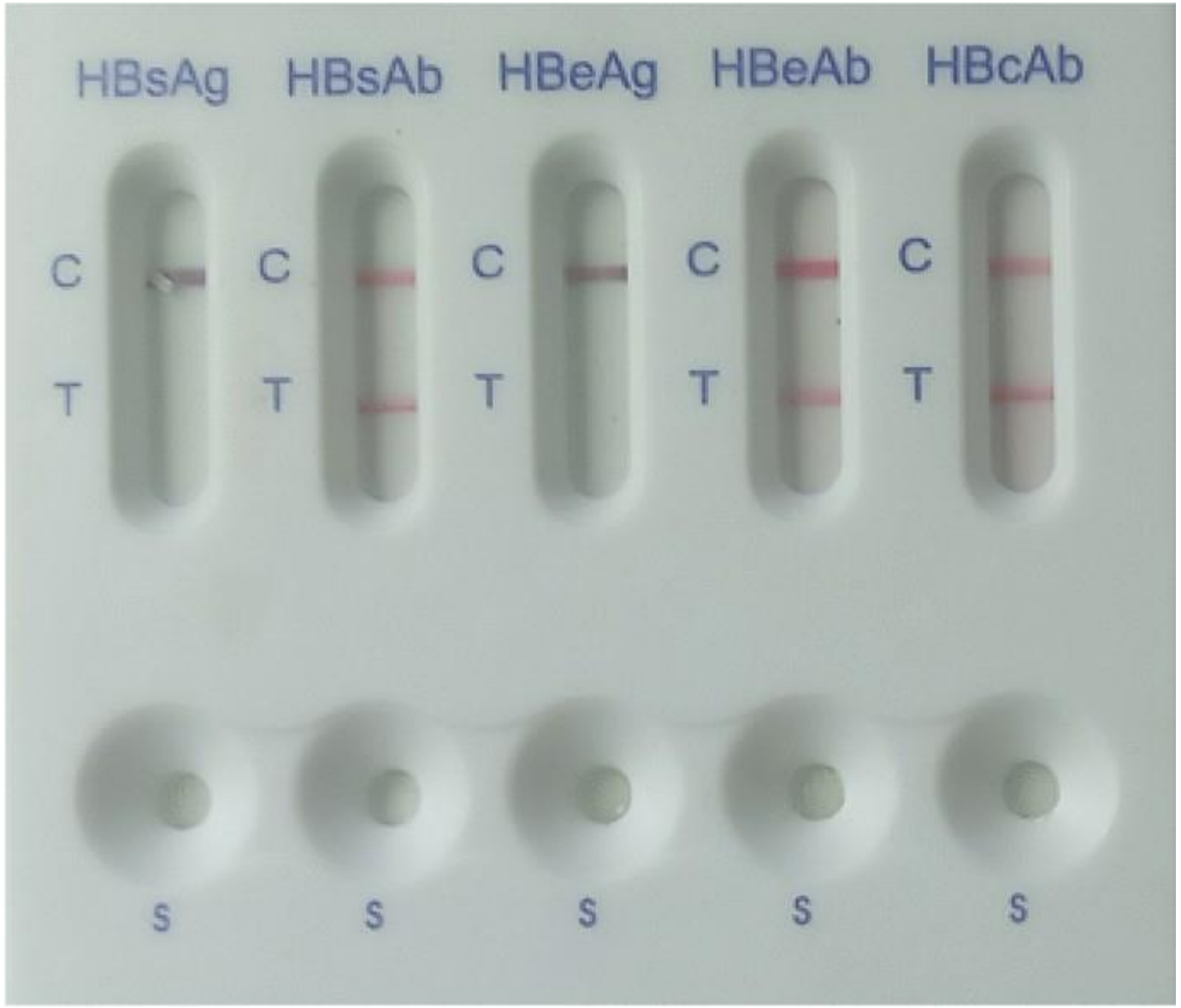
HBV antigens		HBV antibodies	
Hepatitis B surface antigen (HBsAg):	Present in the blood in acute and chronic infections	**HBsAb (IgM then IgG)**	indicates immunity to HBV infection
Hepatitis B core antigen (HBcAg)	Present in hepatocytes but not in blood	**HBcAb**	IgM indicates recent infection and IgG indicates past infection
Hepatitis B envelop antigen (HBeAg)	When present in blood indicates active viral replication and high infectivity.	**HBeAb**	indicates low viral replication

Window period:

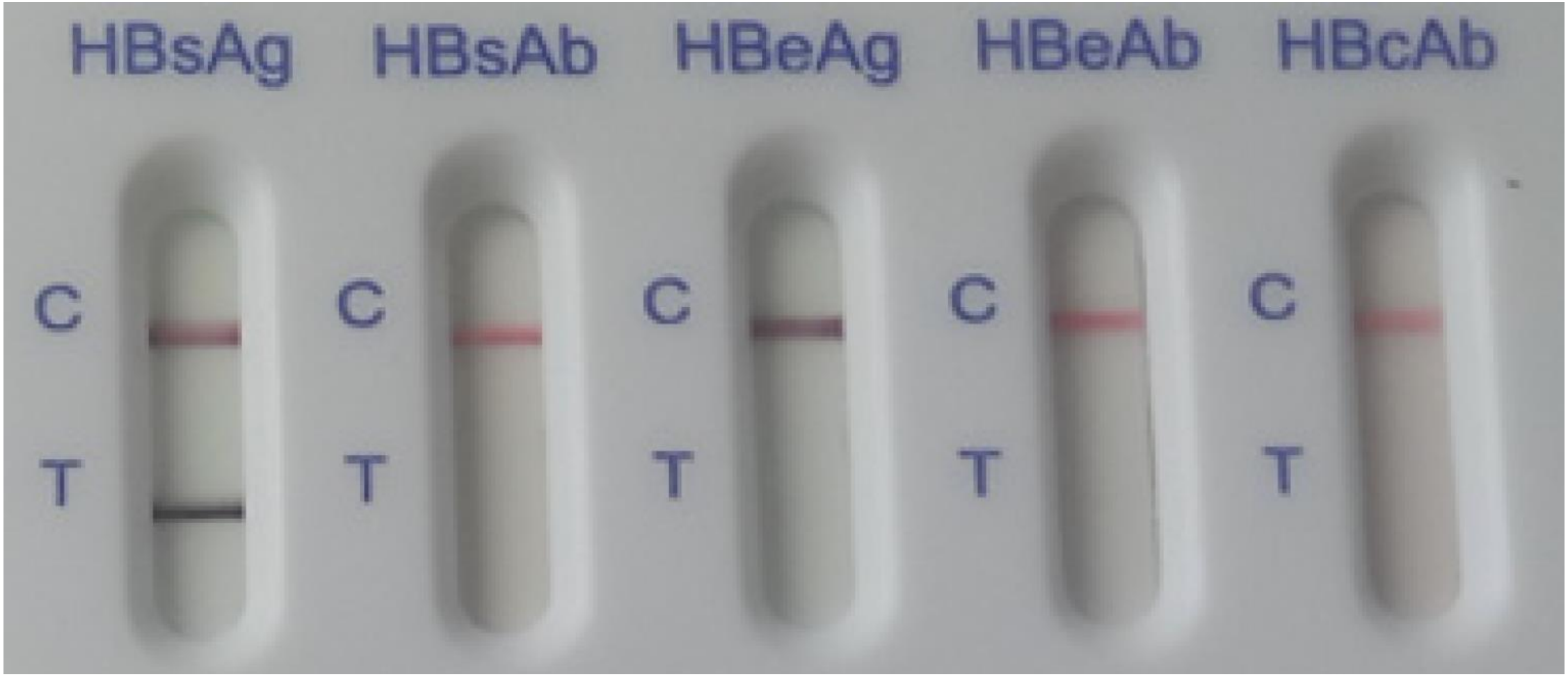
- Period between end of detection of HBsAg and beginning of detection HBsAb.
- The patient is still infectious and the only positive marker is HBc IgM



HBsAg	HBsAb	HBc IgM	HBc IgG	HBeAg	HBeAb	Diagnosis
+	-	+	-	+	-	Acute infection with high infectivity
+	-	-	+	+	-	Chronic infection with high infectivity
+	-	-	+	-	+	Chronic infection with low infectivity
-	-	+	-	-	-	Widow period
-	+	-	+	-	+	Recovered past infection
-	+	-	-	-	-	Vaccinated



What is the diagnosis?



What is the diagnosis?

A. Serologic markers	Results
HBsAg	-
anti-HBc	+
anti-HBs	+

B. Serologic markers	Results
HBsAg	-
anti-HBc	-
anti-HBs	+

C. Serologic markers	Results
HBsAg	+
anti-HBc	+
IgM anti-HBc	+
anti-HBs	-

D. Serologic markers	Results
HBsAg	+
anti-HBc	+
IgM anti- HBc	-
anti-HBs	-

What is the diagnosis?

A……

B……

C……

D……

ANY
QUESTIONS?